Speak **L**ike **A** **M**athematician:

hypotenuse · legs:

Pythagorean triple:

simplest radical form:

Pythagorean Theorem and Its Inequalities

Pythagorean Theorem: in a right triangle, $a^2 + b^2 =$ ________ , where c is the ____________________.

Classifying by the longest side c: $c^2 = a^2 + b^2 \rightarrow$ ____________ triangle · $c^2 > a^2 + b^2 \rightarrow$ ____________ triangle · $c^2 < a^2 + b^2 \rightarrow$ ____________ triangle.

Before classifying, check the ____________________ Inequality: the two shorter sides must sum to MORE than the longest.

Example 1 – Simplify Each Radical (Algebra 1 recall)

a) $\sqrt{72}$ and $\sqrt{45}$

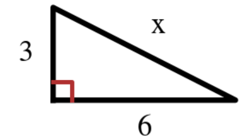
b) $\sqrt{98}$ and $\sqrt{48}$

c) $\sqrt{50}$ and $\sqrt{27}$

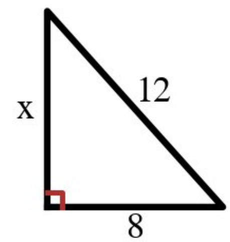
Example 2 – Find x in Simplest Radical Form

Each triangle is at the right.

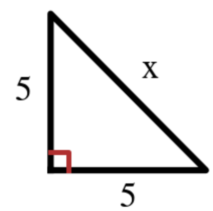
a) Legs 6 and 3.



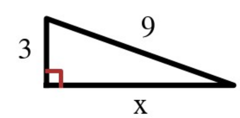
b) Hypotenuse 12, leg 8.



c) Legs 5 and 5.

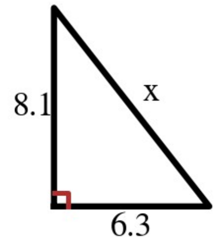


d) Hypotenuse 9, leg 3.

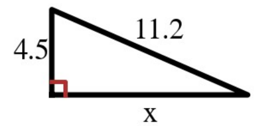


Example 3 – Round to the Nearest Tenth

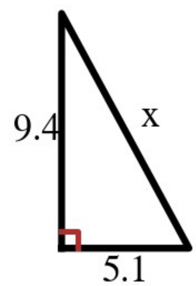
a) Legs 6.3 and 8.1.



b) Hypotenuse 11.2, leg 4.5.



c) Legs 5.1 and 9.4.



Example 4 – Pythagorean Triples

a) Is 20, 21, 29 a triple? Show the check.

b) Is 9, 12, 15 a triple? What smaller triple is it built from?

c) Is 8, 10, 13 a triple?

Example 5 – Classify the Triangle

First check that the sides make a triangle at all; then compare c^2 with $a^2 + b^2$.

a) 6, 8, 9

b) 8, 15, 17

c) 5, 9, 15

d) 7, 9, 13

e) A classmate claims a triangle with sides 1, 1, and 2 is 'right' because $1^2 + 1^2 < 2^2$. What went wrong?